

MATH Unit 4 Question Bank — Discrete Mathematics & Logic

Mathematics (MATH) · Unit 4 · Discrete Mathematics & Logic · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Explain Propositional Logic with a neat diagram and a suitable example.
2. Define Predicate Logic. Derive or justify the key properties involved.
3. Compare Rules of Inference with Set Theory and Relations. When would you prefer one over the other?
4. Define Set Theory and Relations. Derive or justify the key properties involved.
5. Compare Functions with Predicate Logic. When would you prefer one over the other?
6. Write short notes on Mathematical Induction.

Part B — Long answer (10 marks)

1. Define Propositional Logic. Derive or justify the key properties involved.
2. Write short notes on Predicate Logic.
3. Explain Rules of Inference with a neat diagram and a suitable example.
4. Write short notes on Set Theory and Relations.
5. Write short notes on Functions.
6. Write short notes on Mathematical Induction.
7. Explain Propositional Logic with a neat diagram and a suitable example.
8. Write short notes on Predicate Logic.
9. What is meant by Rules of Inference? Discuss its advantages and limitations.
10. What is meant by Set Theory and Relations? Discuss its advantages and limitations.
11. What is meant by Functions? Discuss its advantages and limitations.
12. Define Mathematical Induction. Derive or justify the key properties involved.